

TROUT LAKE, NT, Canada

WMO: 711660

Lat: 60.440N

Lon: 121.240W

Elev: 498

StdP: 95.48

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-40.2	-36.8	-44.8	0.0	-40.1	-40.8	0.1	-36.5	8.1	-8.3	7.2	-7.2	1.1	120	0.720

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	26.1	15.1	24.2	14.6	22.5	13.9	16.7	22.8	15.8	21.5	15.0	20.6	3.4	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.5	11.0	18.7	13.5	10.2	17.9	12.5	9.6	17.1	48.9	22.9	46.2	21.4	43.5	20.6	20.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.4	6.5	5.8	-42.7	29.3	2.3	1.4	-44.4	30.3	-45.7	31.2	-47.1	32.0	-48.8	33.0		
			-42.7	18.4	2.4	1.0	-44.5	19.1	-45.9	19.7	-47.2	20.3	-49.0	21.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.4	-21.7	-16.3	-13.2	-1.1	6.1	12.6	14.9	12.5	7.5	-0.3	-11.8	-18.2
	DBStd	14.18	8.75	7.70	8.68	6.37	4.74	3.37	2.98	3.43	4.02	4.62	7.18	8.51
	HDD10.0	4882	983	737	719	334	137	13	2	15	94	320	654	873
	HDD18.3	7561	1242	971	978	584	381	173	112	183	326	578	904	1131
	CDD10.0	372	0	0	0	1	15	91	154	92	19	1	0	0
	CDD18.3	8	0	0	0	0	0	1	5	2	0	0	0	0
	CDH23.3	252	0	0	0	0	8	65	114	61	4	0	0	0
	CDH26.7	33	0	0	0	0	1	7	14	11	0	0	0	0
Wind	WSAvg	2.5	1.8	2.0	2.4	2.5	2.6	2.6	2.7	2.8	2.9	2.7	2.2	2.0
Precipitation	PrecAvg	410	20	16	16	16	34	63	70	59	41	30	23	21
	PrecMax	521	37	33	33	34	67	112	138	100	101	55	44	42
	PrecMin	266	7	5	1	4	7	14	31	13	14	12	7	9
	PrecStd	59	7	8	9	8	14	26	26	22	18	12	9	7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.1	6.3	10.0	18.3	24.1	27.8	28.6	28.2	23.5	17.1	7.1	8.5
		MCWB	-0.1	1.6	2.8	8.4	11.7	13.2	16.8	16.6	14.0	9.7	3.0	3.0
	2%	DB	-0.9	3.3	6.5	14.4	21.0	25.3	26.1	24.8	20.0	12.1	4.3	3.0
		MCWB	-2.7	-0.4	1.4	6.1	10.6	13.6	16.0	15.5	12.2	6.8	1.0	0.7
	5%	DB	-3.4	0.4	4.1	11.7	18.6	23.1	24.4	22.7	17.7	9.4	1.6	-0.9
		MCWB	-4.8	-2.3	-0.2	4.6	9.6	12.9	15.5	14.5	11.2	4.9	-0.7	-2.8
	10%	DB	-7.3	-2.8	1.3	9.5	15.9	21.1	22.6	20.6	15.4	6.6	-0.9	-4.6
		MCWB	-8.2	-4.5	-1.8	3.5	8.2	12.6	14.7	13.7	9.9	3.4	-2.5	-5.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.0	2.0	3.4	8.8	12.8	16.5	18.4	17.6	14.9	10.5	2.9	1.1
		MCDB	0.8	5.8	9.2	18.2	21.1	22.1	24.1	23.8	20.9	17.0	6.2	3.5
	2%	WB	-3.0	-0.1	1.6	6.5	11.2	15.2	17.2	16.4	13.3	7.2	0.9	-1.6
		MCDB	-1.6	3.4	6.5	14.0	19.4	21.3	23.2	22.4	19.0	11.4	3.5	0.0
	5%	WB	-5.4	-2.1	0.0	5.0	10.1	14.3	16.2	15.3	11.8	5.3	-0.8	-3.6
		MCDB	-4.1	0.5	3.8	10.9	17.6	20.5	22.1	20.8	16.8	8.8	1.4	-1.8
	10%	WB	-9.0	-4.4	-1.6	3.7	8.8	13.3	15.5	14.1	10.3	3.8	-2.7	-6.4
		MCDB	-8.1	-2.7	1.5	9.3	15.7	19.3	21.1	19.3	14.8	6.6	-1.1	-5.4
Mean Daily Temperature Range	5% DB	MDBR	10.9	14.3	15.4	14.3	13.2	12.9	11.9	12.2	11.2	8.6	10.3	10.6
		MCDBR	13.8	18.6	18.5	17.1	18.5	17.5	16.6	17.3	16.3	14.1	11.5	12.6
		MCWBR	12.2	15.2	13.7	10.1	9.8	8.6	8.2	9.3	9.3	8.8	9.8	11.2
	5% WB	MCDBR	13.7	18.6	18.1	16.0	17.1	14.7	14.1	14.9	15.3	13.2	11.4	12.5
		MCWBR	12.4	15.4	13.8	9.8	9.4	7.9	7.7	8.5	9.4	8.5	9.5	11.1
Clear-Sky Solar Irradiance	taub	0.199	0.241	0.261	0.319	0.342	0.343	0.358	0.341	0.301	0.275	0.213	0.171	
	taud	2.184	2.298	2.292	2.281	2.435	2.470	2.445	2.485	2.600	2.534	2.216	2.028	
	Ebn at Noon	621	789	882	879	885	888	863	852	841	750	584	484	
	Edh at Noon	36	61	87	108	100	99	99	88	66	50	35	26	
All-Sky Solar Radiation	RadAvg	0.42	1.34	3.10	4.82	5.56	5.68	5.36	4.34	2.85	1.33	0.48	0.22	
	RadStd	0.03	0.08	0.17	0.27	0.37	0.50	0.35	0.36	0.30	0.13	0.07	0.01	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page